# Core Supervised Learning — Preprocessing, Models & Evaluation (Day 2)

UCI Adult (Census Income) · Same hold-out test set as Day 1 · Due 1 Sept 2026

## 1. Preprocessing Plan

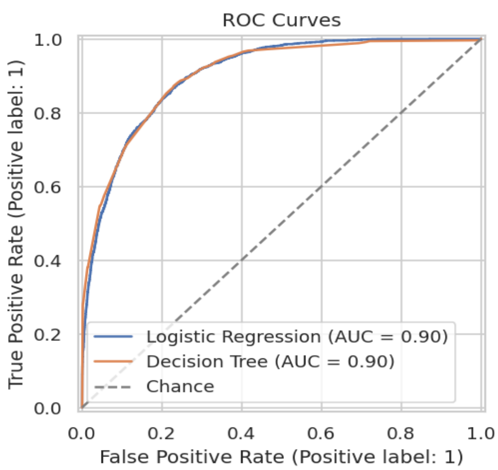
**Numeric (6):** age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week → median imputation + StandardScaler. **Categorical (8):** workclass, education, marital-status, occupation, relationship, race, sex, native-country → most_frequent imputation + OneHotEncoder(handle_unknown='ignore'). Built as a single sklearn ColumnTransformer, fit only on the training set inside each model's Pipeline — preprocessing statistics never see dev or test data, preventing leakage.
**Why median, not mean:** capital-gain/loss are extremely skewed and zero-inflated; mean imputation would inject a distorted, outlier-pulled value. **Why not fill with 0:** 0 is already a real, common value for those columns, so it would hide genuine missingness. **Why OneHotEncoder, not ordinal encoding:** categories like occupation/race have no natural order — ordinal integers would falsely imply ranking, especially damaging for a linear model. **Alternatives considered and skipped:** target/mean encoding (leakage risk without careful cross-fitting) and a constant 'Missing' category for categoricals (kept as a labelled alternative to test on Day 3).

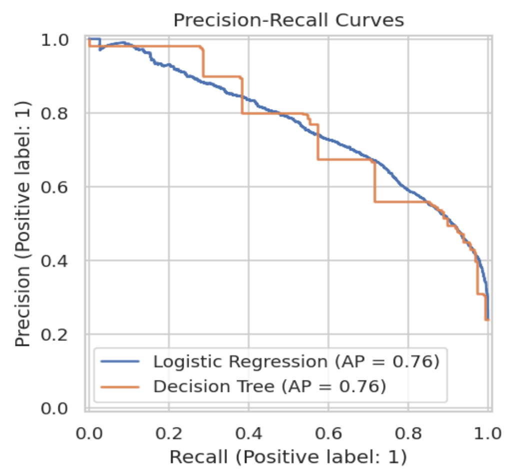
## 2-3. Models & Hold-Out Evaluation

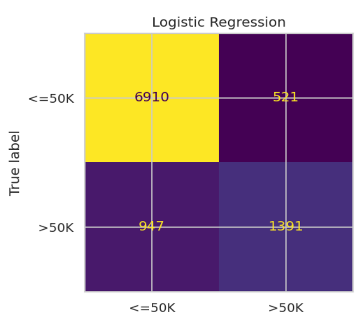
Logistic Regression (liblinear solver, random_state=42) and a Decision Tree (max_depth=8, random_state=42) were each wrapped in preprocessing+estimator Pipelines and fit on the training set only. Both are compared against Day 1's baselines on the identical hold-out test set (n=9,769):

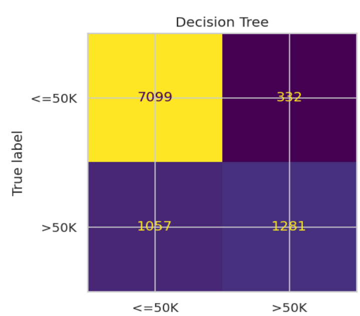
| Model | Accuracy | Precision | Recall | F1 | ROC AUC | PR AUC |
|---|---|---|---|---|---|---|
| Day1: Majority-class | 0.761 | 0.000 | 0.000 | 0.000 | 0.500 | 0.239 |
| Day1: Rule (edu-num>=13) | 0.747 | 0.472 | 0.500 | 0.486 | 0.662 | 0.356 |
| **Day2: Logistic Regression** | **0.850** | **0.728** | **0.595** | **0.655** | **0.904** | **0.760** |
| **Day2: Decision Tree** | **0.858** | **0.794** | **0.548** | **0.648** | **0.904** | **0.758** |

Both models clear the Day 1 rule baseline by a wide margin on precision (our primary metric: +0.26 for Logistic Regression, +0.32 for Decision Tree) and on ROC AUC (~0.90 vs 0.66). **Decision Tree wins on precision (0.794 vs 0.728) but Logistic Regression wins on recall/F1**, and both reach ~0.90 ROC AUC.




**Error type:** both models produce more false negatives than false positives (LogReg: 521 FP / 947 FN; Tree: 332 FP / 1057 FN). Given our precision-first business objective, this works in our favor — both models are conservative about flagging someone >50K, protecting precision at some cost to recall (some genuine high earners go uncontacted, an acceptable trade-off here).

## 4. Interpretability Check

**Logistic Regression — top positive coefficients** (push toward >50K): capital-gain, marital-status=Married-civ-spouse/Married-AF-spouse, relationship=Wife, education-num, and occupation=Exec-managerial — consistent with Day 1's EDA (education, hours, marriage all tracked with higher income). **Top negative coefficients** (push toward <=50K): occupation=Priv-house-serv, education=Preschool, marital-status=Never-married, relationship=Own-child, sex=Female. The negative sex=Female coefficient reflects a real pattern in this historical census data and is worth flagging as a fairness consideration before any real-world deployment, not just a modeling detail.

**Decision Tree — depth & overfitting:** depth=8, 102 leaves, train accuracy=0.860, test accuracy=0.858 — a 0.0025 train-test gap, i.e. **negligible overfitting** at this depth cap. **Top splits:** the tree's very first split is on marital-status=Married-civ-spouse, then on capital-gain, then on education-num — the same three signals Logistic Regression weighted most heavily and the same signals Day 1's EDA flagged as most separating. This convergence across three independent analyses (EDA, linear coefficients, tree splits) is a good sanity check the models are learning genuine signal, not noise.

## 5. Model Selection for Day 3

Both models comfortably beat the Day 1 baselines and are carried forward, for different reasons. **Logistic Regression is the primary candidate** given our precision-focused objective: it is directly interpretable (the coefficient table above can be explained to a non-technical stakeholder), fast to retrain, and its regularization strength (C) plus decision threshold give two easy, well-understood levers to trade recall for precision. **Decision Tree is kept as a secondary candidate** specifically because it already edges out precision (0.794 vs 0.728) by capturing nonlinear interactions (e.g. capital-gain combined with marital-status) that a linear model cannot without explicit interaction terms — if a tuned tree or a later ensemble (Random Forest / Gradient Boosting) clearly and consistently beats tuned Logistic Regression on precision, it becomes the stronger production candidate despite being less directly interpretable.

The ColumnTransformer preprocessing pipeline built in Task 1 is model-agnostic and will be reused unchanged for any new models on Day 3, so future performance differences are attributable to the model, not inconsistent preprocessing.

**Preprocessing changes planned for Day 3:** (1) test the 'Missing'-constant categorical imputation alternative against most_frequent; (2) test dropping fnlwgt (a census sampling weight, not an individual trait, flagged in Day 1); (3) log1p-transform or binarize capital-gain/loss given their extreme skew; (4) bucket the long tail of native-country categories before one-hot encoding; (5) sweep Logistic Regression's C and try class_weight='balanced' for the ~24% class imbalance; (6) tune Decision Tree max_depth/min_samples_leaf against the dev set only. The hold-out test set remains untouched until a final comparison later in the week.